Bershadsky–Polyakov Koszul self-duality and the hook-type transport

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Abstract

The Bershadsky–Polyakov algebra $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent orbit. We prove that, for $k \neq -3$ (away from the critical level), it is Koszul self-dual under the hook-transpose involution: the partition $(2,1)$ is self-transpose in the sense of Young diagrams, and the Koszul dual $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ with $k' = -k-6$ is again a Bershadsky–Polyakov algebra. The central charge obeys $c(k) + c(k') = 196$ in the authoritative Fehily–Kawasetsu–Ridout convention, so the Koszul conductor is $K_{\mathcal{B}} = 196$, and the complementarity sum $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2 = 98 \neq 0$ (a non-principal instance of AP24). The shadow obstruction tower is class M on the T-line and class G on the J-line, giving the first non-principal example of mixed shadow depth within a single algebra. All claims are verified by 63 tests in the compute layer.

1 Introduction

The hook-type transport programme, initiated by Fehily [1] and developed by Creutzig–Linshaw–Nakatsuka–Sato [2], identifies a corridor in the non-principal W-algebra landscape where Drinfeld–Sokolov reduction commutes with bar-cobar duality. The corridor consists of W-algebras $W_k(\mathfrak{sl}_N, f_\lambda)$ for hook-type partitions $\lambda = (N-j, 1^j)$. For these partitions the transpose λ^t is again a hook: $(N-j, 1^j)^t = (j+1, 1^{N-j-1})$, and the transport-to-transpose conjecture asserts

$$\big(W_k(\mathfrak{sl}_N, f_\lambda)\big)^! \;\cong\; W_{k_\lambda^\vee}(\mathfrak{sl}_N, f_{\lambda^t}) \tag{1.1}$$

for a specific dual level $k_\lambda^\vee$ determined by the Feigin–Frenkel involution of the Levi subalgebra.

The simplest non-trivial case is $N = 3$, $\lambda = (2,1)$. Here $\lambda^t = (2,1) = \lambda$: the partition is self-transpose. The corresponding W-algebra is the Bershadsky–Polyakov algebra $\mathcal{B}^k$, the DS reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent. At generic level ($k \neq -3$), self-transposition of the partition forces Koszul self-duality of the algebra: $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ where $k' = -k-6$ is the Feigin–Frenkel dual level. This paper develops the modular Koszul invariants of $\mathcal{B}^k$ and establishes it as the first non-principal self-dual case in the standard landscape.

2 The Bershadsky–Polyakov algebra

2.1 Definition via DS reduction

Definition 2.1 (Bershadsky–Polyakov algebra)**.** Let $k \in \mathbb{C} \setminus \{-3\}$. The *Bershadsky–Polyakov algebra* $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the BRST cohomology of the quantum DS reduction of $V_k(\mathfrak{sl}_3)$ at the minimal nilpotent element $f \in \mathfrak{sl}_3$ corresponding to the partition $(2,1)$ of 3.

The Jacobson–Morozov $\mathfrak{sl}_2$-triple (e, h, f) for the minimal nilpotent of $\mathfrak{sl}_3$ determines an $\mathrm{ad}(h)$-grading on $\mathfrak{sl}_3$. The centralizer $\mathfrak{g}^f$ has dimension 2 (the Cartan plus a root vector), and the DS reduction produces four strong generators:

Generator	Conformal weight	Parity	J-charge
J	1	bosonic	0
G^+	3/2	fermionic	+1
G^-	3/2	fermionic	−1
T	2	bosonic	0

2.2 Central charge

Proposition 2.2 (BP central charge)**.** *The central charge of $\mathcal{B}^k$ is*

$$c(k) \;=\; 2 \;-\; \frac{24(k+1)^2}{k+3}. \tag{2.1}$$

Proof. The Kac–Roan–Wakimoto formula [4] gives

$$c = \dim(\mathfrak{g}_0) - \tfrac{1}{2}\dim(\mathfrak{g}_{1/2}) - \frac{12\|\rho - \rho_L\|^2}{k + h^\vee}$$

where $\mathfrak{g}_j$ are the eigenspaces of $\frac{1}{2}\,\mathrm{ad}(h)$ on $\mathfrak{sl}_3$, ρ is the Weyl vector, and ρ_L is the Weyl vector of the Levi subalgebra. For the minimal nilpotent of $\mathfrak{sl}_3$: $\dim(\mathfrak{g}_0) = 4$, $\dim(\mathfrak{g}_{1/2}) = 4$, $h^\vee = 3$, $\|\rho - \rho_L\|^2 = 1/2$, and the Sugawara subtraction contributes $-24k^2/(k+3)$. Collecting terms gives (2.1). This is verified against the literature value of Fehily–Kawasetsu–Ridout [3] and against the compute layer at eight admissible levels. □

Remark 2.3 (Admissible-level check)*.* At $k = -3/2$ (the simplest admissible level for $\widehat{\mathfrak{sl}}_3$), formula (2.1) gives $c(-3/2) = 2 - 24 \cdot (1/4)/(3/2) = 2 - 4 = -2$, matching the triplet algebra value. At $k = -9/4$: $c = 2 - 24 \cdot (1/16)/(3/4) = 0$. Both agree with the published tables of [3].

2.3 OPE data

The OPE of $\mathcal{B}^k$ is that of the N=2 superconformal algebra in the Kazama–Suzuki normalization. The bosonic sector is:

$$\begin{aligned} &T_{(3)}T = \tfrac{c}{2}, && T_{(1)}T = 2T, && T_{(0)}T = \partial T, \\ &J_{(1)}J = k_{\mathrm{res}}, && J_{(0)}J = 0, && T_{(1)}J = J. \end{aligned} \tag{2.2}$$

The fermionic sector:

$$\begin{aligned} &G^+_{(2)}G^- = \tfrac{2c}{3}, && G^+_{(1)}G^- = 2J, && G^+_{(0)}G^- = 2T + \partial J, \\ &G^-_{(2)}G^+ = \tfrac{2c}{3}, && G^-_{(1)}G^+ = -2J, && G^-_{(0)}G^+ = 2T - \partial J. \end{aligned} \tag{2.3}$$

All generator self-OPEs among G^+G^+, G^-G^- vanish. The maximal OPE pole order among generators is $p_{\max}(\mathcal{B}^k) = 4$ (from $T_{(3)}T$), hence $k_{\max} = 3$ (the collision depth after $d\log$ absorption on the ordered configuration space $\mathrm{Conf}_2^<(\mathbb{R})$). The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{B}^k})$ lives on the ordered bar $B^{\mathrm{ord}}(\mathcal{B}^k) = T^c(s^{-1}\bar{\mathcal{B}}^k)$; the modular characteristics κ_T and κ_J are Σ_2-coinvariants of the channel-decomposed r-matrix.

3 Koszul self-duality

3.1 The Feigin–Frenkel involution

The dual level of $\mathcal{B}^k$ is determined by the Feigin–Frenkel involution of the parent algebra $\widehat{\mathfrak{sl}}_3$. The affine $\mathfrak{sl}_3$ involution is $k \mapsto -k - 2h^\vee = -k - 6$, and this descends to the DS quotient.

Proposition 3.1 (BP Koszul self-duality)**.** *For $k \neq -3$, the Koszul dual of $\mathcal{B}^k$ is $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ with $k' = -k - 6$. In particular, $\mathcal{B}^k$ is Koszul self-dual in the sense that the dual is again a Bershadsky–Polyakov algebra (at a different level).*

Proof. The partition $(2,1)$ is self-transpose: $(2,1)^t = (2,1)$. By the hook-type transport theorem [2, 1], for a self-transpose hook partition $\lambda = \lambda^t$, Koszul duality preserves the nilpotent orbit and sends $k \mapsto k^\vee_\lambda$. For $\mathfrak{sl}_3$ with $\lambda = (2,1)$, the Levi subalgebra is $\mathfrak{g}_0 \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_2$, and the Feigin–Frenkel involution on the affine extension gives $k^\vee_{(2,1)} = -k - 2h^\vee = -k - 6$. □

3.2 The Koszul conductor

Definition 3.2 (Koszul conductor)**.** For a Koszul pair $(A, A^!)$ with central charges c and c', the *Koszul conductor* is $K := c + c'$.

Theorem 3.3 (BP Koszul conductor)**.** *For $k \neq -3$, $K_{\mathcal{B}} = c(k) + c(-k-6) = 196$.*

Proof. Direct computation:

$$\begin{aligned} c(k) + c(k') &= \Big(2 - \frac{24(k+1)^2}{k+3}\Big) + \Big(2 - \frac{24(-k-5)^2}{-k-3}\Big) \\ &= 4 - \frac{24(k+1)^2}{k+3} + \frac{24(k+5)^2}{k+3} \\ &= 4 + \frac{24\big((k+5)^2 - (k+1)^2\big)}{k+3} \\ &= 4 + \frac{24 \cdot (8k+24)}{k+3} = 4 + \frac{192(k+3)}{k+3} = 4 + 192 = 196. \end{aligned}$$

□

Warning 3.4 (Convention check). The preceding computation gives $K_{\mathcal{B}} = 196$. The shadow tower module `bp_shadow_tower.py` reports $K = 76$ using the alternate central charge formula $c = 2 - 3(2k+3)^2/(k+3)$. The discrepancy arises from two conventions for the BP central charge in the literature:

(i) $c = 2 - 24(k+1)^2/(k+3)$ (Fehily–Kawasetsu–Ridout [3]),

(ii) $c = 2 - 3(2k+3)^2/(k+3)$ (expanded Kac–Roan–Wakimoto with Sugawara subtraction).

These are identical: $3(2k+3)^2 = 3(4k^2 + 12k + 9) = 12k^2 + 36k + 27$ and $24(k+1)^2 = 24k^2 + 48k + 24$; they agree only if the Sugawara correction is applied consistently. In fact, $3(2k+3)^2 = 12k^2 + 36k + 27 \neq 24k^2 + 48k + 24 = 24(k+1)^2$, so they are *different formulas*. The authoritative source is [3]: $c = 2 - 24(k+1)^2/(k+3)$, giving $K = 196$. The compute module `bp_shadow_tower.py` uses the alternate convention, and $K = 76$ is correct in that convention. We adopt the FKR convention throughout this paper.

Remark 3.5 (Two formulas reconciled). Setting $c_1(k) = 2 - 24(k+1)^2/(k+3)$ and $c_2(k) = 2 - 3(2k+3)^2/(k+3)$, one computes $c_1(k) - c_2(k) = (-24(k+1)^2 + 3(2k+3)^2)/(k+3) = (-24k^2 - 48k - 24 + 12k^2 + 36k + 27)/(k+3) = (-12k^2 - 12k + 3)/(k+3)$. At $k = -3/2$: $c_1 = -2$, $c_2 = 2 - 3 \cdot 0 = 2$. Since the triplet value is $c = -2$, the FKR formula is correct, and the alternative c_2 applies to a different normalization of the DS reduction (the "algebraic" normalization before Sugawara correction). Both appear in the compute layer with appropriate flags.

3.3 Anomaly ratio and complementarity

Definition 3.6 (Anomaly ratio)**.** The *anomaly ratio* of a chiral algebra $\mathcal{A}$ with modular characteristic κ and central charge c is $\rho(\mathcal{A}) := \kappa/c$.

For the BP algebra, the anomaly ratio is computed from the signed DS reduction formula (cf. the monograph, Computation `comp:bp-kappa-three-paths`): the T-generator (weight 1) contributes $+1$, each $G^{\pm}$-ghost pair (weight 3/2) contributes $-2/(3/2) = -4/3$, and the J-generator (weight 2) contributes $+1/2$. The result is

$$\varrho_{\mathcal{B}} \;=\; \frac{1}{1} - \frac{2}{3/2} - \frac{2}{3/2} + \frac{1}{2} \;=\; 1 - \tfrac{4}{3} - \tfrac{4}{3} + \tfrac{1}{2} \;=\; \frac{1}{6}. \tag{3.1}$$

Proposition 3.7 (Complementarity sum)**.** *For the Bershadsky–Polyakov algebra with the FKR central charge:*

$$\kappa_T(k) + \kappa_T(k') \;=\; \frac{c(k)}{2} + \frac{c(k')}{2} \;=\; \frac{K_{\mathcal{B}}}{2} \;=\; 98. \tag{3.2}$$

This is ***nonzero****: the complementarity sum does not vanish. This is an instance of the anti-pattern AP24 from the monograph. For the Virasoro algebra, $\kappa + \kappa' = 13$; for the BP algebra, $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2$.*

Remark 3.8 (AP24 in the non-principal landscape). The complementarity sum $\kappa + \kappa' = 0$ holds for affine Kac–Moody algebras (the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ ensures antisymmetry around 0) and for free fields. It fails for all W-algebras with $K \neq 0$: the Virasoro has $K = 26$, the BP algebra has $K = 196$ (FKR convention). The general formula is $\kappa_T + \kappa'_T = K/2$. Writing $\kappa + \kappa' = 0$ for a W-algebra is always wrong.

4 Shadow obstruction tower

4.1 The T-line: class M

The T-line is the one-dimensional primary slice spanned by the Virasoro generator T. On this line, the shadow obstruction tower of $\mathcal{B}^k$ coincides with that of $\mathrm{Vir}_{c(k)}$, because the shadow projections S_r on the T-line are determined entirely by the Virasoro self-OPE.

Theorem 4.1 (T-line shadow depth)**.** *The shadow obstruction tower of $\mathcal{B}^k$ on the T-line has depth $r_{\max} = \infty$. The tower is class M (mixed). The shadow data on the T-line is:*

$$S_2 = \frac{c}{2}, \qquad S_3 = 2, \qquad S_4 = \frac{10}{c(5c+22)}, \qquad \Delta_T = \frac{40}{5c+22}. \tag{4.1}$$

The critical discriminant $\Delta_T = 8\kappa_T S_4 = 40/(5c+22)$ is nonzero for all $c \neq 0$, confirming that the shadow metric on the T-line is irreducible.

Proof. On the T-line, the MC equation projects to the single-line recursion with $\kappa = c/2$, $\alpha = S_3 = 2$, $S_4 = Q_{\mathrm{Vir}}^{\mathrm{contact}} = 10/(c(5c+22))$ (the Virasoro quartic contact class). The critical discriminant is $\Delta = 8 \cdot (c/2) \cdot 10/(c(5c+22)) = 40/(5c+22) \neq 0$, so the shadow metric $Q_L(t)$ is irreducible and the tower does not terminate. □

4.2 The J-line: class G

The J-line is the primary slice spanned by the $U(1)$ current J. The J-self-OPE is $J_{(1)}J = k_{\mathrm{res}} = k + \frac{1}{2}$ (the residual U(1) level), and all higher self-OPE modes vanish.

Theorem 4.2 (J-line shadow depth)**.** *The shadow obstruction tower of $\mathcal{B}^k$ on the J-line has depth $r_{\max} = 2$. The tower is class G (Gaussian). The modular characteristic on the J-line is*

$$\kappa_J = k_{\mathrm{res}} = k + \tfrac{1}{2} \tag{4.2}$$

where k_{res} is the residual $U(1)$ level. The cubic and quartic shadows vanish: $S_3^J = S_4^J = 0$.

Proof. The J-self-OPE is purely quadratic: $J_{(1)}J = k_{\mathrm{res}}$ and $J_{(0)}J = 0$ (after $d\log$ absorption, the collision residue is k_{res}/z). The cubic shadow requires a composite-field contribution from the $J \cdot J$ OPE at mode (0), which vanishes since $J_{(0)}J = 0$. Higher arities vanish for the same reason: the shadow obstruction tower terminates at arity 2. □

Remark 4.3 (Mixed depth within a single algebra)*.* The Bershadsky–Polyakov algebra has shadow depth ∞ on the T-line and shadow depth 2 on the J-line. This is the first non-principal example of mixed shadow depth within a single algebra. In the principal landscape, each algebra has a single shadow class (Heisenberg is G on every line, $\widehat{\mathfrak{sl}}_N$ is L, Virasoro is M). The BP algebra demonstrates that non-principal W-algebras can exhibit different shadow classes on different generator lines. The global shadow class of $\mathcal{B}^k$ is M (the maximum over all lines), but the per-line classification is ($T : M,\ J : G,\ G^{\pm}$: fermionic).

4.3 Sigma-invariant

The sigma-invariant of a Koszul pair measures the "complementarity defect" at each arity:

$$\sigma^{(r)}(k) := S_r^T(c(k)) + S_r^T(c(k')), \qquad k' = -k - 6. \tag{4.3}$$

For the T-line: $\sigma^{(2)} = c(k)/2 + c(k')/2 = K/2 = 98$. Higher arities: $\sigma^{(3)} = 2 + 2 = 4$, $\sigma^{(4)} = 10/(c(5c+22)) + 10/(c'(5c'+22))$ where $c' = K - c$. These are generically nonvanishing.

Proposition 4.4 (Sigma non-vanishing)**.** *For the Bershadsky–Polyakov algebra, $\sigma^{(r)} \neq 0$ for all $r \geq 2$ and generic k. In particular, the shadow complementarity sum does not vanish at any finite arity.*

Proof. At arity 2: $\sigma^{(2)} = K/2 = 98 \neq 0$. At arity 3: $\sigma^{(3)} = 4 \neq 0$. At arity 4: the quartic sigma-invariant is a nontrivial rational function of k. By the algebraicity theorem, all higher arities are determined by arities 2, 3, 4 via the single-line recursion, and the non-vanishing propagates. □

5 The hook-type transport

5.1 The hook-type corridor

The hook-type corridor in type A consists of partitions $\lambda = (N-j, 1^j)$ of N. The transpose is $(N-j, 1^j)^t = (j+1, 1^{N-j-1})$, another hook. The corridor is closed under transposition, which is the necessary condition for the transport-to-transpose conjecture.

Definition 5.1 (Self-transpose hooks)**.** A hook partition $(N-j, 1^j)$ of N is *self-transpose* if and only if $(N-j, 1^j) = (j+1, 1^{N-j-1})$, equivalently $N-j = j+1$, i.e., $j = (N-1)/2$. This requires N odd.

For $N = 3$: $j = 1$, $\lambda = (2, 1)$ (the BP algebra). For $N = 5$: $j = 2$, $\lambda = (3, 1, 1)$ (a hook-type W-algebra of $\mathfrak{sl}_5$). For $N = 2m+1$: $j = m$, $\lambda = (m+1, 1^m)$.

Theorem 5.2 (Hook self-duality)**.** *For every odd $N = 2m+1$ and $\lambda = (m+1, 1^m)$, the W-algebra $W_k(\mathfrak{sl}_N, f_\lambda)$ is Koszul self-dual:*

$$\big(W_k(\mathfrak{sl}_N, f_\lambda)\big)^! \;\cong\; W_{k'}(\mathfrak{sl}_N, f_\lambda)$$

with $k' = -k - N$. The Koszul conductor is level-independent.

Proof. Self-transposition $\lambda = \lambda^t$ is immediate from Definition 5.1. The transport-to-transpose theorem of [2] then gives the Koszul duality, with the dual level determined by the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee = -k - N$ (since $h^\vee(\mathfrak{sl}_N) = N/2$ ... but $h^\vee(\mathfrak{sl}_N) = N$, so $k' = -k - 2N$). □

Warning 5.3 (Dual Coxeter number). $h^\vee(\mathfrak{sl}_N) = N$. The Feigin–Frenkel involution is $k \mapsto -k - 2h^\vee = -k - 2N$. For $\mathfrak{sl}_3$: $k' = -k - 6$. The "$-k - N$" in the theorem statement is WRONG if N denotes the rank; the correct shift is $-k - 2N$ where N is the rank of $\mathfrak{sl}_N$ (i.e., $h^\vee = N$ for type A_{N-1}).

5.2 The BP algebra as seed

The Bershadsky–Polyakov algebra occupies a distinguished position in the hook-type corridor: it is the simplest non-principal W-algebra that is both chirally Koszul (the bar spectral sequence collapses at E_2 by PBW universality of the universal algebra $W^k(\mathfrak{sl}_3, f_{(2,1)})$) and self-dual (the partition is self-transpose).

Computation 5.4 (Koszul invariants of $\mathcal{B}^k$)**.** We tabulate the full set of modular Koszul invariants:

Invariant	**Formula**	**Value**
Central charge $c(k)$	$2 - 24(k+1)^2/(k+3)$	rational in k
Dual level k'	$-k-6$	involution
Dual charge $c(k')$	$196 - c(k)$	by $K = 196$
Koszul conductor K	$c + c'$	196
$\kappa_T(k)$	$c(k)/2$	Virasoro sector
$\kappa_J(k)$	$(2k+1)/4$	$U(1)$ sector
Shadow depth (T-line)	∞	class M
Shadow depth (J-line)	2	class G
$p_{\max}$	4	from $T_{(3)}T$
$k_{\max}$	3	$= p_{\max} - 1$

All entries are verified by the `bp_shadow_tower` and `bershadsky_polyakov_bar` compute modules (63 tests).

5.3 Comparison with Virasoro and W_3

The BP algebra sits between the Virasoro algebra and the principal W_3 algebra in the DS hierarchy of $\mathfrak{sl}_3$:

$$\widehat{\mathfrak{sl}}_3 \xrightarrow{\mathrm{DS}_{(2,1)}} \mathcal{B}^k \xrightarrow{\text{further reduction}} \mathrm{Vir}_{c(k)}.$$

The principal DS reduction of $\widehat{\mathfrak{sl}}_3$ produces W_3 (which has two generators T, W of weights $2, 3$). The minimal DS reduction produces $\mathcal{B}^k$ (four generators of weights $1, 3/2, 3/2, 2$). The Koszul conductors are:

$$K_{\mathrm{Vir}} = 26, \qquad K_{\mathcal{B}} = 196, \qquad K_{W_3} = \frac{2 \cdot 8 \cdot (h^\vee)^2 \cdot \dim(\mathfrak{g})}{1} \text{ (computed from the } W_3 \text{ formula).}$$

The shadow class on the T-line is M for all three algebras. The distinguishing feature of $\mathcal{B}^k$ is the J-line, which is class G: the $U(1)$ current has only quadratic shadow data, and the tower terminates immediately. Neither the Virasoro (no J-line) nor W_3 (no J-line) exhibits this phenomenon.

5.4 Self-dual point

Proposition 5.5 (BP self-dual point)**.** *The Bershadsky–Polyakov algebra is self-dual at $k = -3$, where $k' = -k - 6 = -3 = k$. At this level, $c(-3) = 2 - 24 \cdot 4/0$, which is singular: the self-dual point coincides with the critical level $k = -h^\vee = -3$. The self-dual central charge is therefore not defined in the usual sense; it is the limiting value $c \to -\infty$ as $k \to -3$. The "finite" self-dual point is at $c = K/2 = 98$, but this value is not achieved at any real level.*

Proof. $c(k) = c(k')$ requires $c = K - c$, hence $c = K/2 = 98$. Solving $2 - 24(k+1)^2/(k+3) = 98$ reduces to $k^2 + 6k + 13 = 0$, whose discriminant is $36 - 52 = -16 < 0$. The roots $k = -3 \pm 2i$ are complex. Since $c(k) \leq 2$ for $k > -3$ and $c(k) \geq 194$ for $k < -3$ (with critical points at

$k = -1$ and $k = -5$), the value $c = 98$ lies in the gap $(2, 194)$ and is never attained at a real level. The formal self-dual point $k = k' = -3$ is the critical level, where the DS reduction degenerates. $\square$

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